Mudit Jain

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.89
Vivekanand Public Sr Sec School, Bharatpur	Class XII (Board of Secondary Education Rajasthan)	2019	82.40%
Vivekanand Public Sr Sec School, Bharatpur	Class X (Board of Secondary Education Rajasthan)	2017	86.00%

MASTER'S THESIS PROJECT

Bead Profile Prediction and Toolpath Planning for Non-planar Metal AM

Guide: Prof. Venkata Reddy N

Present Work:-

- Developing a mathematical model to predict bead-area asymmetry on planar WAAM substrates via force balance of gravity, arc, plasma drag, and surface tension forces under varied substrate inclination, torch orientation, and deposition direction.
- Formulating a MATLAB regression model to reconstruct bead profiles using bead width, height, and cross-sectional area.

Future Work:-

- Integrating and analyzing the influence of substrate inclination, torch orientation, deposition direction, welding current, and torch travel velocity on bead geometry via force balance to identify stable deposition conditions and improved surface finish.
- Validating the proposed models for deposition on non-planar substrates fabricated using incremental sheet metal forming.

ACADEMIC PROJECTS

Nonlinear Stress Concentration Analysis of a Hyperelastic Plate

FEM Course Project */ Instructor: Prof. Viswanath Chinthapenta*

- Built a 2D plane-stress Abaqus FEA model using the Arruda-Boyce hyperelastic law to study stress around elliptical cutouts.
- Discretized a 1×1 m structural plate with 0.5×0.25 m elliptical cutouts under displacement-driven tensile loading conditions.
- Implemented CPS4R reduced-integration elements with a refined global mesh of 0.01 m to capture high stress gradients.
- Computed a peak stress concentration factor of 9.79, demonstrating significant stress amplification near the elliptical cutouts.

Impact Load Analysis and Structural Optimization of a Go-Kart Chassis

B.Tech Project */ Guide: Prof. Sandeep Kumar Parashar*

- Modeled a circular tubular chassis in SolidWorks having a 1500 mm wheelbase and 900 mm track width for torsional rigidity.
- Calculated a 19.16 kN impact load from vehicle mass, velocity change, and impact duration, and applied it to the chassis.
- Performed structural FEA for front, rear, and side impact load cases to identify critical members and deformation zones.
- Added a chassis cross-stiffening member to redistribute impact loads, achieving a ~ 22% reduction in overall

deformation.

INTERNSHIP EXPERIENCE

Tata Motors Commercial Vehicles | Driveline Systems & Maintenance

B.Tech Summer Internship

- Gained practical exposure to maintenance and repair of commercial vehicles, focusing on clutches, gearboxes, and brake units.
- Witnessed differential and transmission teardown and repair of assemblies to understand gear contact, wear, and failures.
- Explored the role of leaf spring suspension linkages in supporting chassis loads and contribution to suspension performance.

TECHNICAL SKILLS

- **CAD &Drafting Tools:** AutoCAD (2D Engineering Drawings), SolidWorks (3D Parts, Assemblies, Weldments).
- **CAE & Numerical Tools:** Abaqus (Nonlinear FEA), ANSYS Fluent (Fluid Flow), MATLAB (Numerical Modeling).
- **Manufacturing Processes:** Incremental Sheet Metal Forming, Wire Arc AM (WAAM), FDM 3D Printing, CNC.

KEY COURSES

Fundamentals of Additive Manufacturing | Design for Metal Additive Manufacturing | Design for Manufacturing and Assembly | Finite Element Method | Computational Fluid Dynamics Tools | Advanced Material Joining Processes

EXTRA-CIRCULAR ACTIVITIES

- **Social Media Coordinator | Indian Knowledge System (IKS) Student Chapter | IIT Hyderabad:-** Coordinated social media content, event promotions, and digital outreach to enhance student engagement.

- **Hobbies:-** Cycling, Cooking, Mobile Games (Mini-Militia)